

Sensor Selection

Temperature Sensor:

Want Sensors for:

Inside incubator: chamber floor and lid

incubator reservoir

outside incubator

objective temp

Inside Incubator Sensing:

Ideally want an accuracy around 0.1 -0.3 deg celsius

Resistance temperature detectors are regarded as the most accurate, so I will first explore these as an option.

^^ okay so I've seen now that there are different classes of these they aren't all super accurate... there are classes:

- Class AA (~1/3 DIN): $\pm(0.10 + 0.0017 \times t)^\circ\text{C}$ → about $\pm 0.16^\circ\text{C}$ at 37°C
- Class A: $\pm(0.15 + 0.002 \times t)^\circ\text{C}$ → about $\pm 0.22^\circ\text{C}$ at 37°C
- Class B: $\pm(0.30 + 0.005 \times t)^\circ\text{C}$ → about $\pm 0.49^\circ\text{C}$ at 37°C

^^ also see I need a resistance to digital convertor like: MAX31865 for these sensors

^^ - **2-wire**: cheapest, but both leads' resistance gets added straight into the reading. On a PT100 with ~1.5m of cable, that's often 0.5–1°C of error — enough to wipe out a Class A element's accuracy.

- **3-wire**: a third wire lets the converter estimate and subtract the lead resistance (assuming both current leads match), removing most of that error. This is the standard for most probes.
- **4-wire**: a separate pair senses voltage with almost no current flowing, so lead resistance barely matters at all — the most accurate option, but needs a converter that supports it.

^^ there are also PT1000 probes where often they use two wire: PT100 but built to be 1000Ω at 0°C instead of 100Ω — 10x the nominal resistance, and 10x the sensitivity (~3.85Ω per °C instead of 0.385Ω per °C). That same 0.25–0.4Ω of lead resistance that would cost a PT100 roughly 0.6–1°C only costs a PT1000 about 0.07–0.1°C — small enough that 2-wire becomes genuinely reasonable at this cable length. That's exactly why manufacturers pair 2-wire wiring with PT1000 elements far more often than with PT100

^^ another consideration is the probe length if it is too long it's going to be problematic in the incubators

^^ need to decide if need one or two of these

RTD Options

1. Micro-robotics: PT100 RTD 0.5% accuracy 1m length, **R110**, but it doesn't specify class so this one is a bit suspicious, seems to be 3 wire
<https://www.robotics.org.za/PT100-3P-05-1M?search=resistance%20temperature%20detect&description=true>
2. RS online: PT1000 RTD Class A, 2 wire, probe length 5cm, **R377**, <https://za.rs-online.com/web/p/rtd-sensors/8968399> (around $\pm 0.23-0.28^{\circ}\text{C}$, $0.29-0.39^{\circ}\text{C}$ worst-case)
3. RS online - PT100 RTD Class A, 3 wire, probe length 10cm, **R489**, <https://za.rs-online.com/web/p/rtd-sensors/3730372>
^^ probably too long
4. RS online - PT100 RTD Class A, 3 wire, probe length 60cm, **R1194**, <https://za.rs-online.com/web/p/rtd-sensors/4573704> **$\pm 0.224^{\circ}\text{C}$** .
^^ a bit more accurate but 3x the price

Resistance to Digital Converters

^^^ thinking for my incubator temp sensor i just want to use a dedicated ADC specifically for this so that it is accurate

1. DigiKey- Adafruit PT1000 RTD Temperature Sensor Amplifier MAX31865, **R245**,
https://www.digikey.co.za/en/products/detail/adafruit-industries-llc/3648/7807995?gclsrc=aw.ds&gad_source=1&gad_campaignid=18332725675&gbraid=0AAAAADrbLlgWVdCk7N2T1gKv6viUufx7L&gclid=CjwKCAjws_DTBhB_EiwAXZknGddQx48g0Uij_H0imSFVlzDMjwfEw_4huPOcol8hAhagLx8jyGxjwRoCTQQQAvD_BwE
2. DigiKey - Adafruit MAX31865 RTD PT100 Amplifier, **R245**,
https://www.digikey.co.za/en/products/detail/adafruit-industries-llc/3328/6562952?gclsrc=aw.ds&gad_source=1&gad_campaignid=18332725675&gbraid=0AAAAADrbLlgWVdCk7N2T1gKv6viUufx7L&gclid=CjwKCAjws_DTBhB_EiwAXZknGTj9Ddtl4DFL6XBQGX1HjdPz0s2amku3C0CqBxKjVwpHpDVvDM4AUROc_OkQAvD_BwE
^^ good data sheet, pretty sure can be used for both above: <https://cdn-learn.adafruit.com/downloads/pdf/adafruit-max31865-rtd-pt100-amplifier.pdf>

^^ I've started new research and it actually doesn't seem like this is the way to go, these come up as super accurate but they are very large (doesn't make sense for a small incubator)... Having one for calibration purposes would be good but not as a main sensor
^^ can use as the ground truth sorta thing

Outside, Reservoir

for these sensors we don't need them to be as accurate just need to know when to heat the water and can have the external temp as a disturbance variable for the incubator temp loop

Sensor Options

1. RS online- DFRobot Waterproof DS18B20 Digital Temperature Sensor for Arduino Temperature Sensor Sensor Board for DS18B20 Arduino, says $\pm 0.5^\circ\text{C}$ accuracy, **R145**, https://za.rs-online.com/web/p/sensor-development-tools/2049893?cm_mmc=ZA-PLA-DS3A-_-google-_-PLA_ZA_EN_PMAX_Catch+All-_-_-2049893&matchtype=&&gclid=aw.ds&gad_source=1&gad_campaignid=20574379828&gbraid=0AAAAADkeWNOmvitSNFkp67kkmGUA7nNrl&gclid=CjwKCAjws_DTBhB_EiwAXZknGXfy01p80QHXXKIM7UbcjRtd7EWMzhpZ1dGBRx935sVAfZ-QaVyXNBoCytCQAvD_BwE
 ^^ they also have this on takealot: https://www.takealot.com/waterproof-ds18b20-digital-temperature-sensor-for-arduino/PLID56355677?gclid=aw.ds&gad_source=1&gad_campaignid=23333443430&gbraid=0AAAAADuiBExxe4d86x2_4bC3CDfVEaqo-&gclid=CjwKCAjws_DTBhB_EiwAXZknGWpD6HNY28-StZm6eDVSX3AaZAoknSfc5go4TyJDCB5T_niyQ0hnARoC6QoQAvD_BwE
 ^^ this one is ideal, doesn't need an ADC
 ^^ data sheet on RS is useful

so one thing i've discovered is that these temp sensors are large and this isn't going to work for some of the temp sensing, for example for some of the incubators they have a thing thermistor that they stick into the wells for measuring temp (get an accurate idea of how the wells are doing, i also like the idea of this for testing), then also for the glass they use a thermistor at the edge, I'm thinking what I will do is use the PT100 to calibrate these so that they are more accurate... would be good to have more options)

Well and Glass

"precision glass bead NTC thermistor probe small" or "interchangeable precision thermistor immersion probe" — look for ones specifying tight tolerance ($\pm 0.1^\circ\text{C}$ or $\pm 0.2^\circ\text{C}$ interchangeability) and a small glass-bead or epoxy tip in the 1–2mm range.

Options

1. Digikey - THERMISTOR NTC 10KOHM 3976K BEAD, **R245** - https://www.digikey.co.za/en/products/detail/te-connectivity-measurement-specialties/GA10K3MCD1/5277239?gclid=aw.ds&gad_source=1&gad_campaignid=20198569094&gbraid=0AAAAADrbLlggfCDof4_cMN9jyJID7loDA&gclid=Cj0KCQjw4orUBhCjARIsAlbF3qyz6VWnBQJwfekMfLptg2C1TqJspU6kq_gSz5Zq-b2ahVTkaC1fDKUaAkCoEALw_wcB
2. DigiKey - THERMISTOR NTC 10KOHM 3478K BEAD - **R65** - https://www.digikey.co.za/en/products/detail/tdk/B57541G1103F000/3712554?gclid=aw.ds&gad_source=1&gad_campaignid=18332725675&gbraid=0AAAAADrbLlg5ytnlC_OmD_SiQ-yiDdlzM&gclid=Cj0KCQjw4orUBhCjARIsAlbF3qwcUR9Of39aqGAto1gxpVcxVD7l4LsTG82_oR2FVqZ9jOMw1RW4MmlaAuFNEALw_wcB
- 3.

okay the # GA10K3MCD1 is best and actually probably just makes the most sense to use this for all the temps, instead of having many different ones and calibrations making life tricky

ADC for thermistors

if going to have thermistors x4 makes sense to use ADC (only need one I2C line then to my STM don't want to take up 4 of my STM pins)

1. Digikey - ADS1115 16BIT ADC 4CH PROG GAIN - **R245** -

https://www.digikey.co.za/en/products/detail/adafruit-industries-llc/1085/5761229?gclid=aw.ds&gad_source=1&gad_campaignid=18332725675&gbraid=0AAAAADrbLIg5ytnIC_OmD_SiQ-yiDdlzM&gclid=Cj0KCQjw4orUBhCjARIsAlbF3qxS74ZGaUo_P6JQXCBFgdwsLzVNQ3WGVpTo3ddop01dBTldRiffGF4aAge7EALw_wcB

^^ - **16-bit resolution** — way more than you need in theory, but it means each ADC "step" is tiny, so your thermistor's inherent nonlinearity (not the ADC) becomes your accuracy bottleneck, not resolution. Good for slow, precision thermal measurement like yours.

- **4 single-ended (or 2 differential) inputs per chip** — covers lid, well, water, air on one board
- **I2C, 2 pins, cheap breakout boards** (~\$3-5, widely available — Adafruit's is the well-documented reference design, but generic clones work fine)
- **Built-in PGA (programmable gain amplifier)** — lets you scale the ADC's input range to match your divider's actual voltage swing at 37°C, which improves effective resolution in the exact range you care about rather than wasting bits on the full 0-VCC range
- **Slow sample rate is a non-issue for you** — its max is ~860 samples/sec, absurdly fast for something that changes over seconds/minutes thermally. You'll run it much slower than its ceiling.
- **Well-supported on STM32** — plenty of existing HAL/I2C driver examples and libraries, so you're not writing an I2C driver from scratch

One alternative worth knowing about: MCP3424 — 18-bit, also I2C, similar idea, sometimes preferred for slightly better differential noise rejection, but it's pricier and less commonly used in hobbyist/student projects so you'll find fewer examples/community support. ADS1115 is the more practical choice for you given time constraints on a thesis.

Carbon Dioxide Sensor

^^ one thing I have noticed is that the state of the art options have a sample side where they pump out gas from the incubator to test the CO2 levels and they change the input based on this.. seems like a better solution

^^ major issue is that high humidity is not good for these sensors, condensation will make

them inaccurate so this is a major concern that's why an external sample output makes sense here

^^ needs to be one rated for humidity

^^ now i see some of them can handle the humidity so may not necessarily have to have the external pump/sample thing

pretty much every CO2 sensor in lit review used an NDIR sensor so this is clearly the way to go

Quick summary of CO2 sensor considerations:

- **Type matters:** single-beam NDIR (cheapest, MH-Z19B/SCD30) vs dual-beam NDIR (drift-resistant) vs photoacoustic (SCD4x — smaller/robust, not more accurate)
- **Accuracy:** SCD30 $\pm(30\text{ppm}+3\%)$ > MH-Z19B $\pm(50\text{ppm}+3\%) \approx$ SCD41 $\pm(40\text{ppm}+2.5\text{--}5\%)$
- **No ADC needed** — MH-Z19B (UART) and SCD30/41 (I2C) are self-contained smart sensors, digital output only
- **Critical gotcha for your build:** Automatic Baseline Correction (ABC/ASC) assumes periodic exposure to ~400ppm fresh air to self-zero — your sealed chamber won't get that, so it **must be disabled in firmware** (documented command on both), with manual/periodic calibration instead

More on Accuracy:

normal air is approx 500ppm

5% CO2 in ppm: $5 \times 10,000 = 50,000$ ppm

error = fixed_ppm + (percentage ÷ 100) × reading_in_ppm

- **Great:** under $\pm 1,000$ ppm — matches the tightest commercial systems (Okolab, Tokai Hit)
- **Good:** $\pm 1,000\text{--}2,000$ ppm — this is where Yan's DIY build sits, and it's explicitly described in your own lit review as "within the same range as the commercial systems"
- **Okay:** $\pm 2,000\text{--}5,000$ ppm — still within the commercial range, just at the loose end (Ibidi)
- **Not great:** anything past $\pm 5,000$ ppm — worse than every commercial or DIY system reviewed in your own thesis

NDIR Options:

1. Digi key - SCD30 - SENSOR CARBON DIOXIDE I2C OUTPUT - **R777** -
https://www.digikey.co.za/en/products/detail/sensirion-ag/SCD30/8445334?gclid=Cj0KCQjw4orUBhCjARIsAlbF3qWkuQcnVO5-FnPyqpDPyCQ6HflwEXVnRfZXrV5r1k1Z-PhxNilAV5AaAnfwEALw_wcB

^^ says: Humidity operating conditions Non-condensing. Valid for CO2 sensor. 0 – 95 %RH, can't condense

2. Digi Key- SDC41 - STEMMA QT SCD-41 CO2 HUMID TEMP - **R820** - https://www.digikey.co.za/en/products/detail/adafruit-industries-llc/5190/15194670?gclid=aw.ds&gad_source=1&gad_campaignid=18332725675&gbraid=0AAAAADrbLIg5ytnlC_OmD_SiQ-yiDdlzM&gclid=Cj0KCQjw4orUBhCjARIsAlbF3qySoz5KZ2SodRTZoqvAPETTXPHjdAW97IFr2fUjpshl3EjEN2OnRekaAl9BEALw_wcB
3. Micro- Robotics - # MH-Z19 NDIR CO2 Sensor 5000PPM - **R861** (bigger, more expensive and less accurate than SCD30), <https://www.robotics.org.za/MH-Z19?search=%20MH-Z19B&description=true>

^^ also just realised even having an external sample chamber doesn't solve the condensation issue... so i mean it's fine ratehr just have it inside incubator

Humidity Sensor

so the SCD30 has a humidity sensor onboard so i can just use this for humidity reading